The FUJI Family

Pyramids on flat-ended triangles: from the tetrahedron through FUJI to the flat hexagon, the antiprism theorem, and the members found in tilings, crystals and our sculpture



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Licence CC BY 4.0 · Data, scripts, proofs and checks: studies/2026-10-14-fuji-family at https://github.com/DICTOR-Master/kaleidohedra.
The family, FUJI, the dictospheres, the sculptures and all names are DICTO's. Computation, proofs and checking with Claude (Anthropic).

Abstract.

FUJI is the hexagonal pyramid that closes the corners of our thirteen-icosahedron cluster, the Lotus seed [1]. Its base is a **flat-ended triangle**: a hexagon with all angles 120° and sides 1 and 1/φ alternating. The same shape kept turning up elsewhere, in our cuboctahedral PET-bottle spheres and in the colouring of the Lotus seed, so we asked whether FUJI belongs to a family.

It does. Put an apex over a flat-ended triangle with sides 1 and r, with all six side edges of length 1. This gives a one-parameter family **FUJI(r)**, running from the regular tetrahedron (r = 0) through FUJI (r = 1/φ) to the flat regular hexagon (r = 1). We derive the base angles and prove:

- **The base angles** (derived): cos L = (1 + 2r)/3 and cos² S = (2 + r)/(3(2 − r)).
- **The antiprism rule:** the member whose base continues the plane of the n-gonal antiprism's triangles is **r = 2 cos(π/n) − 1**, so that 1 + r is the short diagonal of the regular n-gon (for n = 3, its side).
- **The wheel bound:** flat wheels crossing along different spokes number at most 3 (square), 4 (hexagonal) or 6 (decagonal), and five octagonal wheels are impossible. The octahedron gives the tetrahedron, the square antiprism √2 − 1, and the pentagonal antiprism (the middle of the icosahedron) gives FUJI.

The members turn up in several places:

- **The golden hexagon** of a construction after Odom is FUJI's base; the same construction with any extension gives every member with r < 1.
- **The space-filling truncated tetrahedron** of Damasceno, Engel and Glotzer has the r = 1/2 member's base as its faces.
- **In a breathing pyrochlore** the kagome hexagons are flat-ended triangles, with r equal to the breathing ratio; as a Wythoff construction it tiles space for every r.
- **The equal-depth cube–octahedron solid** has the cuboctahedral member's base, r = (√3 − 1)/2, as its eight triangles, and its cube corners are exactly that member's apexes; we prove the two conditions are the same.
- **Cleo-Icos and Antonius:** the icosahedron's central pyramid is the corner of the prolate golden rhombohedron, which splits into Cleo-Icos + Antonius + Cleo-Icos; so the two fill space, two to one, and build the rhombic hexecontahedron.

1. Introduction

FUJI was found as a gap piece. In the Lotus seed, twelve icosahedra round a small one, the corners where three icosahedra meet are filled by a hexagonal pyramid with all six side edges 1, which we named after Mount Fuji [1]. It is not made of the Mosseri–Sadoc tiles [Papadopolos and Ogievetsky 1999; Koca et al. 2020] that make its partners AXE and CLEO.

Its base had appeared before. Designing our dictospheres, 3D-printed spheres with 14 PET-bottle ports in a cuboctahedral arrangement, we wanted sharp equilateral triangles round the ports and settled for blunted, flat-ended ones, to keep an even margin round every port. That is where we first met the shape; the spheres themselves are not examples of the family, which is about pyramids, but they led us to the cube–octahedron solid of Section 5.4. Colouring the Lotus seed, we hoped for regular hexagons and found the same flat-ended triangles. Our DeltaZip and B1NR sculptures use rounded triangles, the curved form of the same idea.

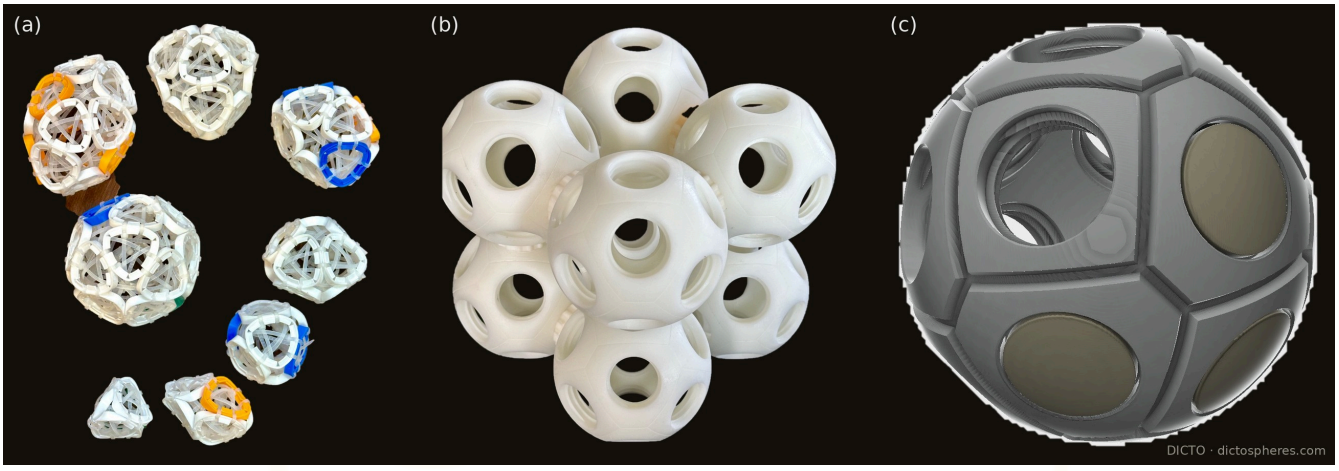


Figure 1. Our sculptures. (a) DeltaZips: deltahedra built from rings tied with zip ties, each ring a rounded triangle. (b) Thirteen dictospheres joined, twelve round one. (c) A dictosphere design, with caps on some of its ports. Photographs and design: DICTO.

The same outline appears at the centre of the DeltaZips logo: the recycling symbol's three arrows make a triangle with its corners taken off.

This paper takes FUJI's shape as the starting point of a family (Section 2), finds its bases in a classical construction (Section 3), proves which member belongs to which antiprism (Section 4), and follows the members into tilings, crystals and our sculpture (Section 5).

2. The family

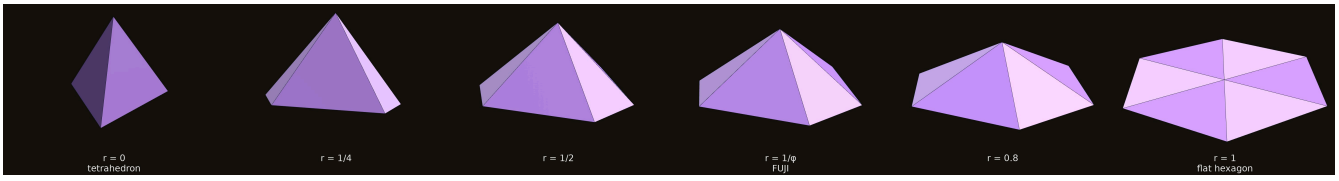


Figure 2. FUJI(r) at one scale, all side edges 1: the tetrahedron (r = 0), r = 1/4, r = 1/2, FUJI (r = 1/φ), r = 0.8 and the flat hexagon (r = 1). Darker faces: the three equilateral triangles; lighter: the three isosceles triangles 1, 1, r.

The base. A flat-ended triangle with sides 1 and r is a big equilateral triangle of side 1 + 2r with an equilateral corner of side r cut off at each corner. In the literature it is the isogonal hexagon or truncated triangle t{3}. All six corners lie on one circle, of radius² (1 + r + r²)/3. So an apex at height h over the centre has all six side edges equal, and they are 1 when

$h^2 = (2 - r - r^2)/3$.

The volume is (√3/12)(1 + 4r + r²) h. The side faces are three equilateral triangles (on the long sides) and three isosceles triangles 1, 1, r (on the short sides), with apex angle 2 arcsin(r/2).

The base angles (proofs in the appendix):

$\cos L = (1 + 2r)/3$ on the long sides, **$\cos^2 S = (2 + r)/(3(2 - r))$** on the short sides.

r	member	height	angle L	angle S	side faces
0	regular tetrahedron	√(2/3)	70.53°	–*	four equilateral
1/4		3/4	60°	49.11°	
1/2		0.6455	48.19°	**41.81°**	
1/φ	**FUJI**	1/√3	**41.81°**	37.38°	isosceles faces are golden triangles (36°)
1	flat	0	0	0	six equilateral triangles in one plane

*At r = 0 the short sides vanish, so S is undefined (the formula's limit, 54.74°, has no face to measure).

Two of the angles are the same angle, 41.81° = arccos(√5/3). In the Lotus seed this is the gap two icosahedra leave either side of a shared edge. It appears as FUJI's long-side angle and as the r = 1/2 member's short-side angle. The formulas show why: $\cos^2 L = 5/9$ exactly at r = 1/φ, and $\cos^2 S = 5/9$ exactly at r = 1/2.

3. The golden hexagon after Odom

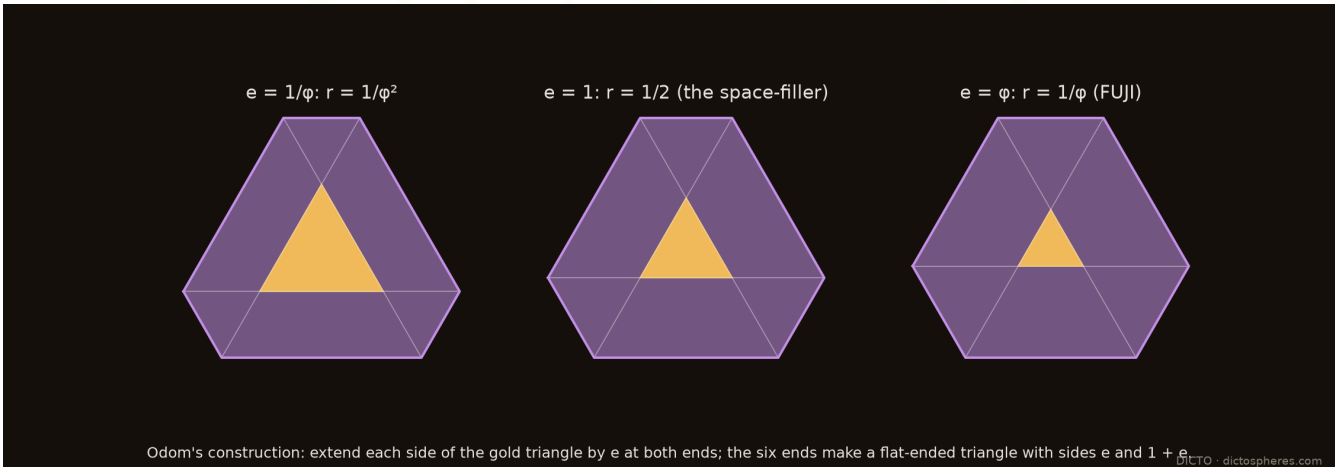


Figure 3. Extend each side of the gold triangle by e at both ends: the six ends make a flat-ended triangle with sides e and 1 + e. With e = φ (right) it is FUJI's base.

George Odom found the golden ratio in the equilateral triangle [Odom and van de Craats 1983]. A related construction, published as the Wolfram Demonstration "Hexagons and the Golden Ratio" after Odom, extends each side of an equilateral triangle by the golden ratio at both ends; the six outer points form a hexagon whose two side lengths are in the golden ratio.

This golden hexagon is FUJI's base. (Odom's own construction, Problem E3007, is a different one: the line through two side midpoints of an equilateral triangle, extended to the circumcircle.)

Rebuilt here for any extension e, the six points always form an equiangular hexagon with sides e and 1 + e: a flat-ended triangle with r = e/(1 + e). So the same construction, with extension e, gives every member with r < 1 (the flat hexagon, r = 1, would need e infinite). This is a change of parameter, not a new result. The extension e = 1 gives r = 1/2, the base of the member in Section 5.2.

4. The antiprism rule

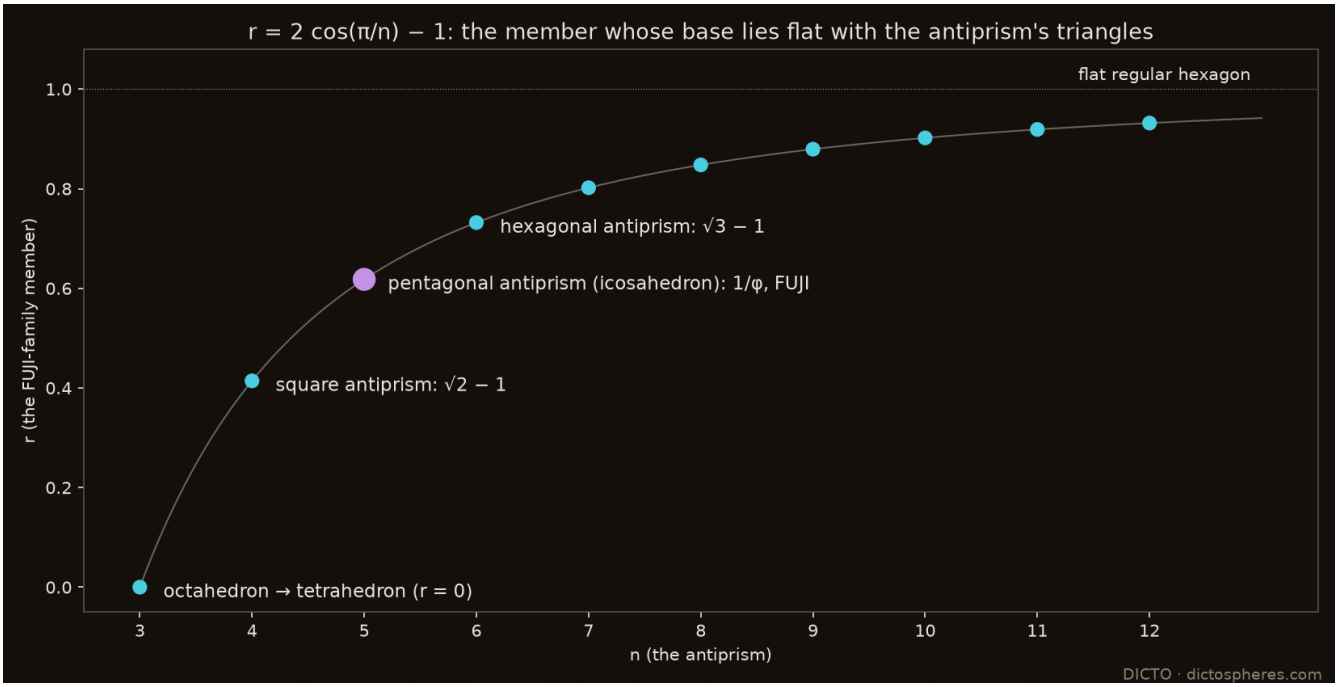


Figure 4. The member that matches each antiprism: r = 2 cos(π/n) − 1. FUJI is the five-fold member.

In the Lotus seed, FUJI's base lies in one plane with the icosahedron triangles on its long sides: 41.81° + 138.19° = 180°. We call this the **face-plane condition**: FUJI(r) is glued by one of its equilateral faces to a unit triangle of a solid X, and its base continues the plane of the face of X across that triangle's edge, so that L + D_X = 180°, where D_X is X's dihedral angle at that edge. The edge is between two triangles for the icosahedron, the octahedron and the antiprisms (their triangle–triangle edges), and between a triangle and a square for the cuboctahedron and the rhombicuboctahedron, where the base continues a square. The condition holds when

r = (−3 cos D_X − 1)/2.

Solid (edge, dihedral angle)	member r
icosahedron (and the pentagonal antiprism, its middle), triangle–triangle (138.19°)	**1/φ: FUJI**
cuboctahedron, triangle to square (125.26°)	**(√3 − 1)/2**
square antiprism, triangle–triangle (127.55°)	**√2 − 1**
rhombicuboctahedron, triangle to square (144.74°)	**(√6 − 1)/2**
octahedron = triangular antiprism, triangle–triangle (109.47°)	0, the tetrahedron

Proposition 1. For the n-gonal antiprism with unit edges, the member meeting the face-plane condition at a triangle–triangle edge is **r = 2 cos(π/n) − 1**. Equivalently, **1 + r = 2 cos(π/n), the short diagonal of the regular n-gon with unit sides** (for n = 3, its side).

The antiprism's dihedral angle, cos D_n = (1 − 4 cos(π/n))/3, is standard (it is listed for single antiprisms in the OEIS); the proposition combines it with the base-angle formula, so the new content is one line of algebra.

Proof (outline; the dihedral computed in full in the appendix). Place the antiprism's corners at T_k = (R cos 2kθ, R sin 2kθ, h) and B_k = (R cos (2k + 1)θ, R sin (2k + 1)θ, −h), with θ = π/n and R = 1/(2 sin θ). The two equilateral triangles on the side edge T_0B_0 have their third corners T_1 and B_{-1}, each √3/2 from the edge's midpoint M. A short computation gives $(T_1 - M)\cdot(B_{-1} - M) = -(4 \cos \theta - 1)/4$. So cos D_n = (1 − 4 cos(π/n))/3, and r = (−3 cos D_n − 1)/2 = 2 cos(π/n) − 1. ∎

The antiprisms therefore index the family by the n-gon's short diagonal:

- n = 3, the octahedron, gives the tetrahedron;
- n = 4 gives the silver ratio's inverse, √2 − 1;
- n = 5, the icosahedron's middle, gives the golden ratio's inverse, FUJI;
- n = 6 gives √3 − 1;
- as n grows, r tends to the flat regular hexagon.

Golden and silver. The members for n = 4 and n = 5 are the inverses of the two best-known metallic means. For n = 5, r = 2 cos 36° − 1 = φ − 1 = 1/φ, the golden ratio's inverse: FUJI. For n = 4, r = 2 cos 45° − 1 = √2 − 1 = 1/(1 + √2), the inverse of the **silver ratio** 1 + √2. Both are short diagonals: 1 + 1/φ = φ, the pentagon's diagonal, and 1 + (√2 − 1) = √2, the square's. So FUJI has a silver sister one step down the same formula, on the square antiprism, and seen down its axis that antiprism's outline is the **regular octagon**, as the pentagonal antiprism's is the regular decagon (Figure 5). The family has square and pentagonal cousins. For k = 3, 4, 5, …, a flat-ended k-gon is a regular k-gon with its corners cut: an equiangular 2k-gon with sides 1 and r alternating. With all side edges 1 the long faces are equilateral, and cos L = (1 + r/cos(π/k))/(√3 tan(π/k)). The pyramid lies flat at r = 2 sin(π/k − π/6), which is 1, 2 sin 15° and 2 sin 6° for k = 3, 4, 5, and 0 or less from k = 6 on, where the regular unit hexagon is already flat. So there are families only for k = 3, 4 and 5. The square family's member on the rhombicuboctahedron (triangle–square edge) is r = 1 − 1/√2. Whether the silver members connect to octagonal quasicrystals (Ammann–Beenker tilings, inflation factor 1 + √2) we leave open.

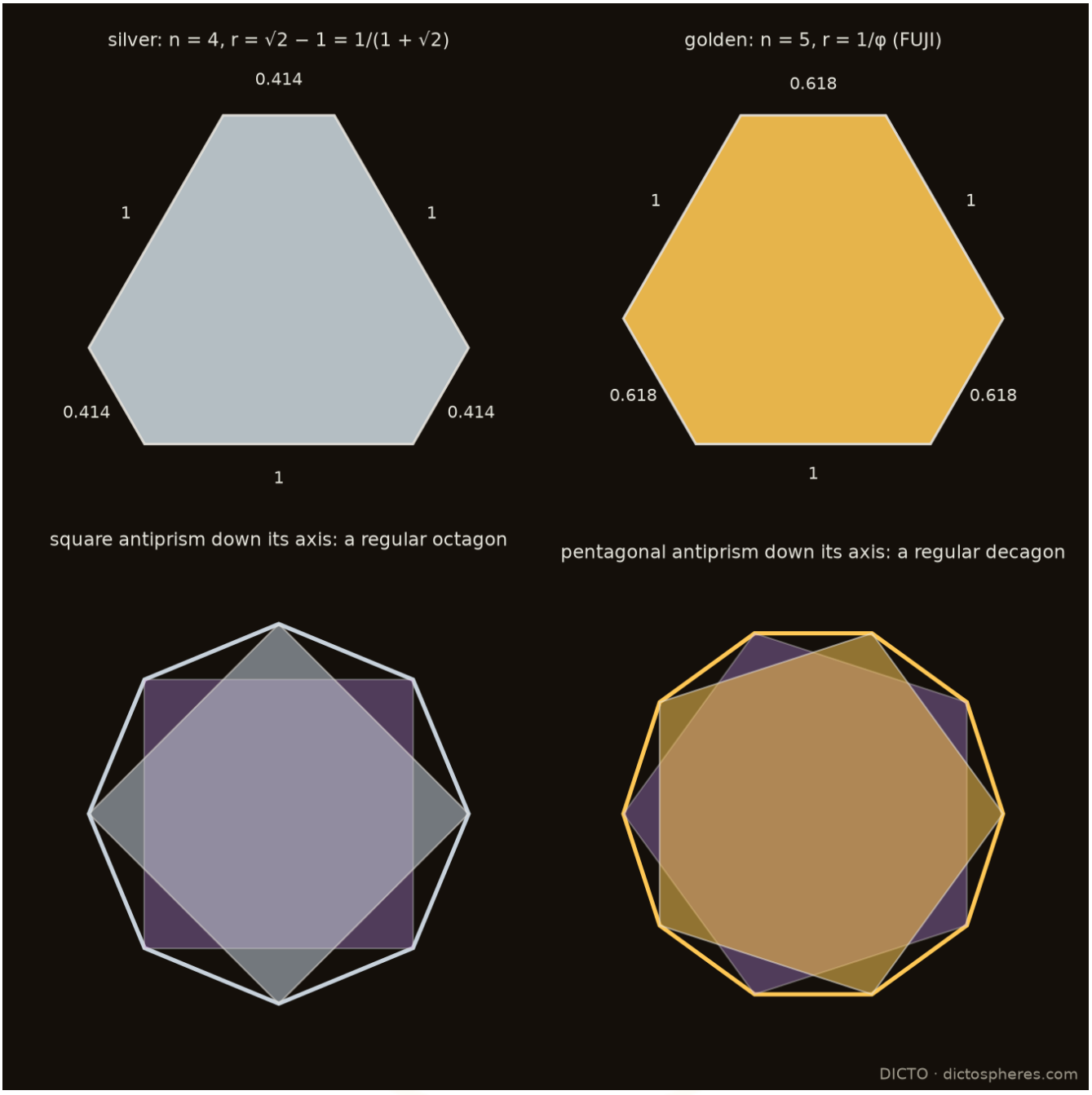


Figure 5. The silver member (n = 4, r = √2 − 1) and the golden member FUJI (n = 5, r = 1/φ): their flat-ended triangles (above), and, separately, their antiprisms seen down the axis, whose outlines are the regular octagon and decagon (below).

5. Where the members live

5.1 FUJI in the Lotus seed

The Lotus seed is a regular icosahedron of edge φ^2, exactly: 12 icosahedra, a small centre, 60 AXE, 20 FUJI and 20 CLEO, or our 63-piece build of a dodecahedron, 12 J11, 30 AXE and 20 FUJI [1]. Each of its 20 big faces is three icosahedron triangles and one FUJI base. FUJI is the five-fold member of the family, which is consistent with its place in the icosahedron (the face-plane condition is necessary, not sufficient, for a cluster to close).

5.2 The space-filling truncated tetrahedron (r = 1/2)

Damasceno, Engel and Glotzer studied truncated tetrahedra from the tetrahedron to the octahedron, and found that the member at truncation t = 1/2 fills space on its own [Damasceno et al. 2012]. Its four hexagons have sides in the ratio 2 : 1 (checked from their published corners): they are the r = 1/2 member's base, the e = 1 hexagon of Section 3.

5.3 Breathing pyrochlores

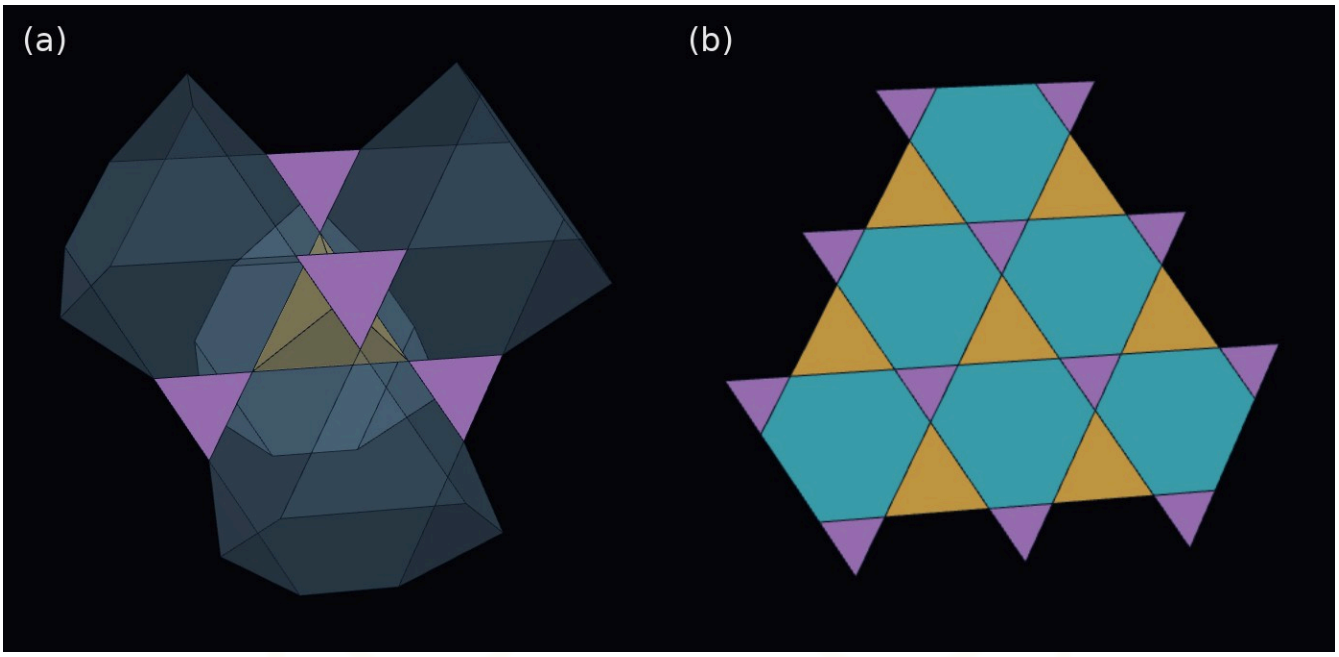


Figure 6. The Golden Breathing Pyrochlore world in the Kaleidohedra app. (a) Big (gold) and small (purple) tetrahedra meet tip to tip, with see-through truncated tetrahedra between them. (b) One kagome layer: every hexagon is FUJI's flat-ended triangle.

A pyrochlore is a network of tetrahedra sharing corners: every corner belongs to one "up" and one "down" tetrahedron, tip to tip. In a **breathing pyrochlore** the two sets have different sizes, edge 1 and edge r. Such materials exist among the chromium spinel oxides [Okamoto et al. 2013]. Every kagome hexagon then has sides 1 and r alternating, with all angles 120°: the flat-ended triangle with the breathing ratio as its r.

The space between the tetrahedra is filled by two kinds of truncated tetrahedra, both with flat-ended-triangle hexagons. The tetrahedra and truncated tetrahedra fill space for every r. This is a Wythoff construction [Coxeter 1973]: in the affine group $\tilde{A}_3$ (whose diagram is a square), ring two adjacent nodes at distances d_0 and d_1; the cells are two tetrahedra, with edges in the ratio $r = d_1/d_0$, and two kinds of truncated tetrahedra (the quarter cubic honeycomb at r = 1). Wythoff's construction gives a tiling for any generating point in the fundamental simplex, not only the uniform one; here the point runs along the edge where the other two mirrors meet, so r takes every value. We also checked volumes, faces and neighbours numerically for r = 0.2, 0.382, 0.5, 0.618, 0.8 and 1. As r tends to 0 the small tetrahedra shrink to points, one kind of truncated tetrahedron becomes an octahedron and the other a tetrahedron, and the crystal becomes the tetrahedral–octahedral honeycomb. The two kinds of truncated tetrahedra share the same hexagon and are twin members of

Damasceno's truncation family, one on the tetrahedron's side and one on the octahedron's. So the breathing pyrochlore is the FUJI family made into a crystal:

- r = 1 is the ordinary pyrochlore;
- r = 1/2 contains the space-filling truncated tetrahedron as one of its cells (the tetrahedron-side kind);
- **r = 1/φ, the golden breathing pyrochlore, has FUJI's base in every hexagon.**

It can be built cell by cell in the Kaleidohedra app, with a slider for the breathing ratio.

5.4 The cube–octahedron solid

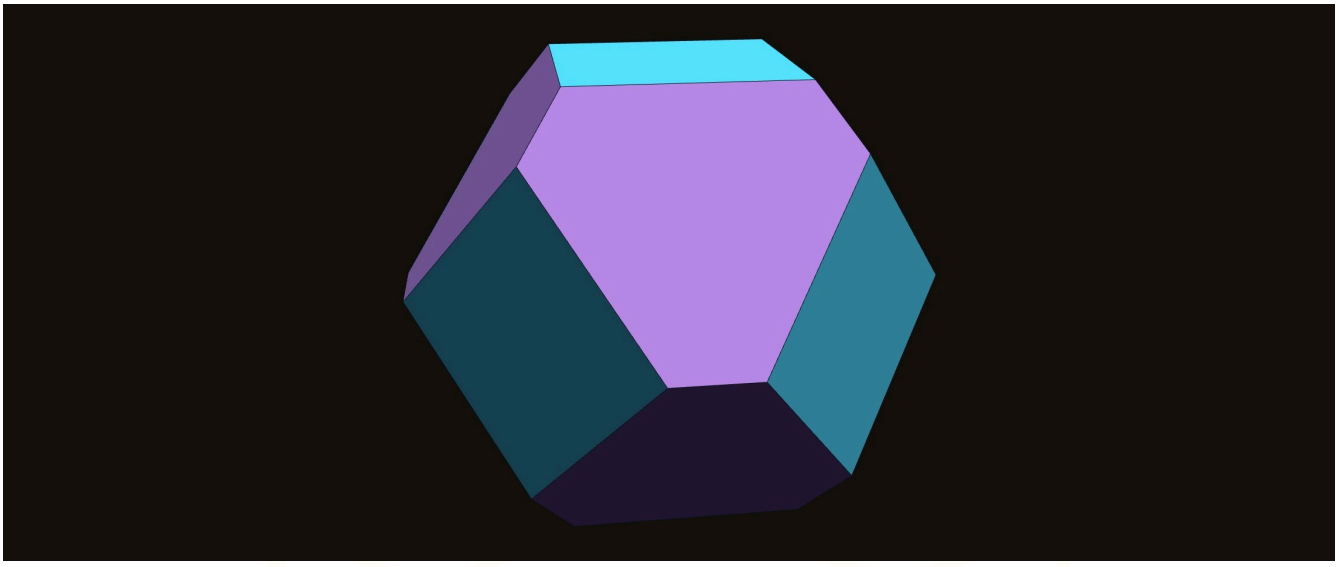


Figure 7. The cube ∩ octahedron with both sets of faces at the same distance from the centre: 6 squares (cyan) and 8 flat-ended triangles (wisteria), r = (√3 − 1)/2, meeting short side to short side.

Cut a cube by an octahedron with both sets of faces at the same distance from the centre ($|x|$, $|y|$, $|z| \le 1$ and $|x| + |y| + |z| \le \sqrt{3}$). The solid has 6 squares and 8 flat-ended triangles, and all 14 faces touch one inner sphere. Its triangles are exactly the **cuboctahedral member, r = (√3 − 1)/2 = 0.366**. (Equal depth is also the rule of even clearance: across an edge shared by two faces, the distances from the two face centres to the edge are equal exactly when the faces lie at the same distance from the centre.)

Why the cuboctahedral member. Take the solid $|x|, |y|, |z| \le 1$, $|x| + |y| + |z| \le S$, and put an apex at the cube's corner (1, 1, 1) over the triangle on $x + y + z = S$. For $3/2 < S < 2$ (where the triangles' long sides face the squares), its six side edges are all $\sqrt{((2 - S)^2 + 1)}$, and its three long faces lie in the cube's faces. Those faces are equilateral exactly when $2(S - 1)^2 = (2 - S)^2 + 1$, that is $S^2 = 3$, which is the equal-depth solid. So **equal depth and the cuboctahedral member are the same condition: at equal depth, the cube's eight corners are exactly the apexes of copies of FUJI((√3 − 1)/2), scaled by √2(√3 − 1) ≈ 1.035, standing on the eight triangles.** The cuboctahedron's member rests on the same angle: in each case a long face and the base lie in a cube plane and an octahedron plane, so L = 54.74° and r = (√3 − 1)/2.

The truncated octahedron is the r = 1 member of the same cube–octahedron family of solids, with regular hexagons. Every member has the same dihedral angles (109.47° and 125.26°), but only the truncated octahedron, whose hexagons are centrally symmetric, fills space by translation (Kelvin's cell). It is to the equal-depth solid what the regular hexagon is to FUJI: the symmetric shape at r = 1.

5.5 Wheels, and the CLEO sphere

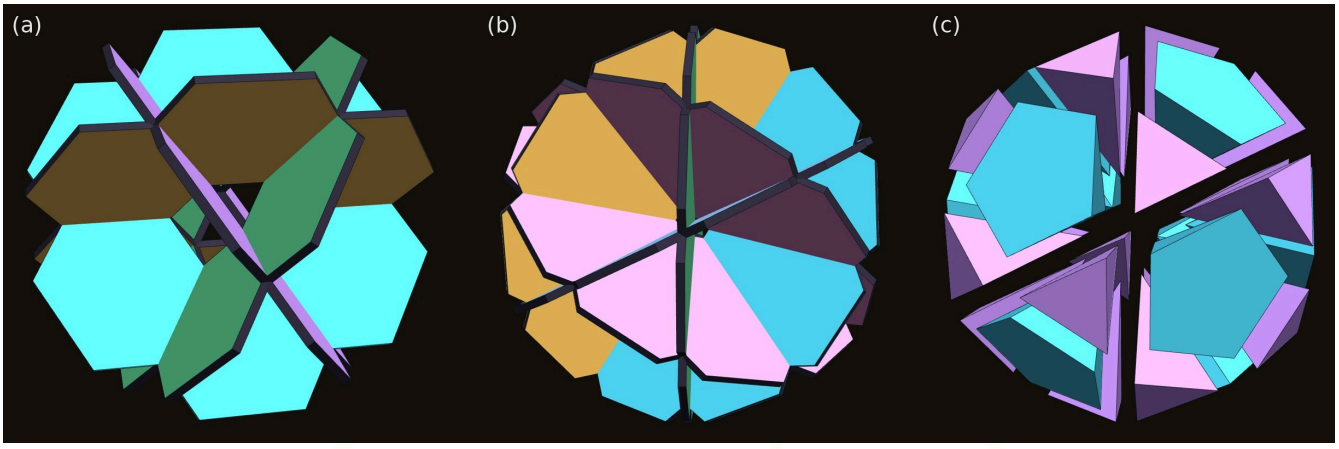


Figure 8. (a) Four hexagonal wheels of flat-ended triangles (drawn with FUJI's, r = 1/φ; any r > 0 works), overlapped into the cuboctahedron's outer structure. (b) Six decagonal wheels of golden triangles, the icosidodecahedron's. (c) The CLEO sphere pulled apart: 20 CLEOs (wisteria) and 12 pentagonal pyramids (cyan), all tips at the centre.

Our DeltaZip wheels led to a construction in the same spirit. A cuboctahedron's 24 edges split into four regular hexagons, one in each of its four central planes; a wheel of six triangles in each plane (centre to rim), with every triangle blunted into a flat-ended triangle, gives four wheels that overlap into the cuboctahedron's outer structure (Figure 8a). Each spoke is shared by two wheels, and the blunted corners keep the hub and the twelve crossings clear. **Four is the most, if every two wheels meet along a different spoke:** a wheel with 2m spokes has m spoke lines, each crossing wheel uses one of them, so n wheels need $n - 1 \le m$. Hexagonal wheels (m = 3) allow at most four, decagonal (m = 5) at most six, square (m = 2) at most three; at the maximum every spoke line lies in exactly two wheels. These maxima are attained by the equatorial great circles of the octahedron, the cuboctahedron and the icosidodecahedron, the classical quasi-regular polyhedra [Coxeter 1973]. (Planes through a single common line meet in any number, but all along the same spoke.) The octagonal case in between is impossible: five such circles would form the one simple arrangement of five great circles (no three through a point), whose triangles, all sides 45°, have angles 65.53°, forcing the central pentagon's angles to 114.47°, while an equilateral spherical pentagon of side 45° with equal angles has angles 122.25°. The icosahedral counterpart has six wheels, in the planes square to the five-fold axes, each a decagon of ten golden triangles (sides φ, φ, 1): the icosidodecahedron's six great decagons (Figure 8b).

The six wheels cut the icosidodecahedron into pieces with their tips at the centre. With edge 1/φ the icosidodecahedron has circumradius exactly 1, so each piece over a triangle has an equilateral base 1/φ and side edges 1: **it is CLEO**. **Twenty CLEOs pointing inwards and twelve pentagonal pyramids** (side edges 1, base edge 1/φ, volume √5/12 each) **fill the icosidodecahedron exactly**, volume $(11\sqrt{5} - 5)/6$ (Figure 8c). It is the icosahedral twin of

the four hexagonal wheels cutting the cuboctahedron into 8 regular tetrahedra and 6 square pyramids.

Each pentagonal pyramid divides about its axis into **five congruent right-angled triangular pyramids**, each with one golden face (1, 1, 1/φ, the same as CLEO's side faces), a base triangle with sides 0.5257, 0.5257 and 1/φ (72° at the centre of the pentagon), and volume √5/60. Equilateral CLEOs themselves cannot fill the pentagonal pockets (a pentagon cannot be tiled by equilateral triangles); these five are the pocket's own CLEO-like pieces. CLEO cannot fill an icosahedron from its centre on its own: its tip angles are 36°, where 20 pieces closing a sphere need 63.4°, so twenty CLEOs cover only about 29% of the directions round the centre. The pentagonal pyramids take the rest.

5.6 Six circles and the icosahedron

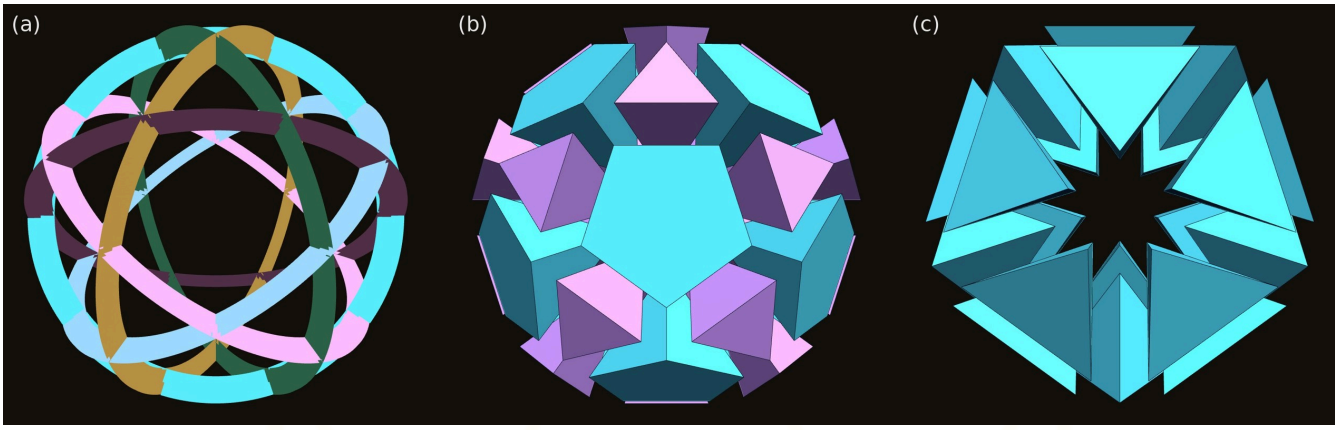


Figure 9. (a) Six equal slotted rings in the planes square to the five-fold axes, seen down a five-fold axis. (b) The CLEO sphere they frame, seen the same way: 20 CLEOs in the triangular windows, 12 pentagonal pockets. (c) Faceted flat across the pentagon centres: 20 Cleo-Icos filling an icosahedron.

How many crossed circles make an icosahedron? Four make the cuboctahedron, and four is the most that meet in that way (Section 5.5). **Six** circles, one in each plane square to a five-fold axis, cross every 36° round each circle and divide the sphere into **20 triangles and 12 pentagons**, the icosidodecahedron's pattern. The icosahedron is there, though the circles never draw its edges: its **12 corners are the centres of the 12 pentagonal windows**, and its **20 faces lie over the 20 triangular windows**. Six equal flat rings slotted together, alternately from the outside and the inside at each crossing, are all identical (one template, cut six), which is how our six-ring DeltaZip balls go together. An alternating slot pattern exists for any set of great circles with no three through one point, since their crossings then form a four-valent map (as in alternating knot diagrams); here the equal crossing angles (63.43°) and equal spacing (36°) make the six rings identical.

The pieces complete the route. The triangular windows hold the 20 CLEOs pointing inwards; each pentagonal window holds five Penta-cleo (Section 5.5). Each CLEO with the three Penta-cleo on its golden sides fills the part, inside the icosidodecahedron, of one face-cone of the icosahedron whose corners are the pentagon centres. Cutting flat across the three pentagon centres removes the parts outside that icosahedron (CLEO's outer corners are at radius 1, beyond the icosahedron's corners at 0.851), and leaves the **Cleo-Icos**: the icosahedron's central pyramid on one face, tip at the centre and flat base outside, side edges 0.851 and base

edge 2/√5 = 0.894. Twenty of them make the icosahedron, as the central pyramids of any solid make that solid; the point is the route, not the dissection. The Cleo-Icos itself is well known under other names: it is the unit tetrahedron of the Mackay icosahedron, the shells of equal spheres round a centre that most icosahedral metal clusters and nanoparticles follow (edge ratio 1/sin 72° = 1.0515 and 63.43° at the centre; Mackay 1962), the slightly strained tetrahedron of icosahedral multiply twinned particles (Ino 1969; Marks 1984), and the "golden tetrahedron" of Kabai's twenty-tetrahedra cluster (2014). Regular tetrahedra cannot close the icosahedron (five round an edge leave 7.36°); the Cleo-Icos is the shape that does. What we add is its place on our route and, in Section 5.7, its partners. Six circles, then, lead to the icosahedron in a roundabout way: through the icosidodecahedron and the CLEO sphere.

The icosahedron, filled or not, still does not fill space (its dihedral angle 138.19° leaves 41.81° gaps), and the twenty pieces alone do not either: their dihedral angles are 72° along the side edges (five close round a line through the centre) and 69.09° along the base edges, and no combination that includes a base edge, with 72° angles and flat 180° faces, makes 360°. In space they need a partner piece: Antonius (Section 5.7).

5.7 Cleo-Icos and Antonius fill space

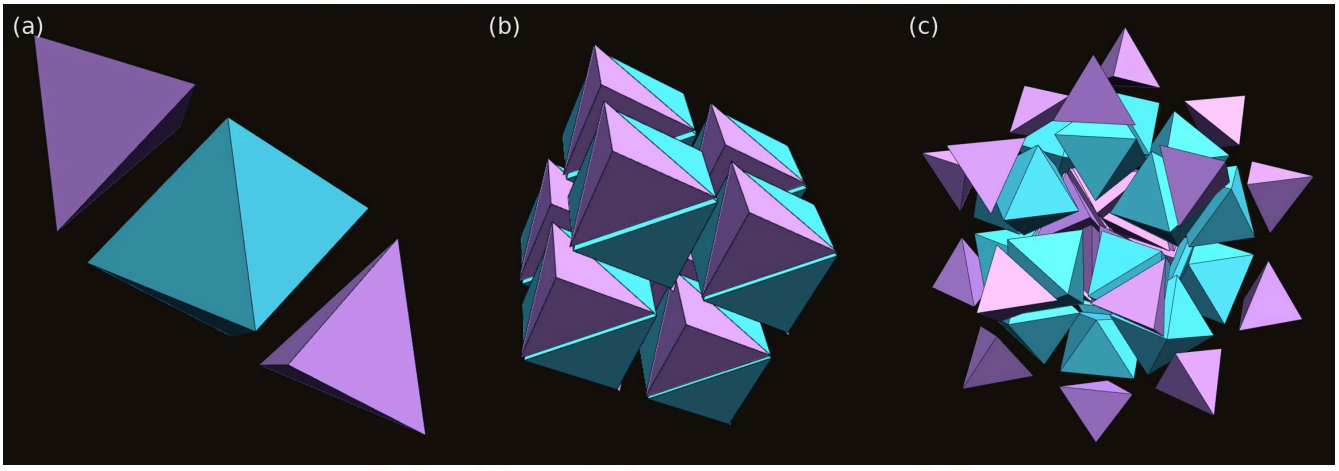


Figure 10. (a) One prolate golden rhombohedron = Cleo-Icos + Antonius + Cleo-Icos, pulled apart along its long axis. (b) Eight rhombohedra, 16 Cleo-Icos (wisteria) and 8 Antonius (cyan), a piece of the space-filling. (c) The rhombic hexecontahedron: 20 Cleo-Icos round the centre (the icosahedron), 20 Antonius, and 20 Cleo-Icos at the tips, pulled apart.

The Cleo-Icos cannot fill space alone (Section 5.6), but it can with one partner. The three edges from the icosahedron's centre to the corners of one face, u, v and w, meet at arccos(1/√5) = 63.43°, the face angle of the **prolate golden rhombohedron**. So u, v and w span a golden rhombohedron (edge 0.851, dihedral angles 72° and 108°), and the Cleo-Icos is exactly its corner at one sharp tip. In coordinates along u, v and w the rhombohedron is 0 ≤ x, y, z ≤ 1, and the planes x + y + z = 1 and x + y + z = 2 cut it into three pieces: a Cleo-Icos at each sharp tip, and between them **Antonius**, a convex trigonal antiprism with two equilateral faces (edge 2/√5) and six side faces identical to the Cleo-Icos's (0.851, 0.851, 0.894). Antonius has exactly **four times the Cleo-Icos's volume**, and the rhombohedron six times.

Golden rhombohedra fill space in a lattice, and every rhombus face is cut along the same diagonal from both sides. So **Cleo-Icos and Antonius fill space, two to one, face to face,**

with no mirror images. Along Antonius's equilateral edges its dihedral angle is 110.91°, which with the Cleo-Icos's 69.09° makes 180°: the faces meet flat. We checked a patch of 2,187 pieces: no gaps or overlaps, every face matched, every edge and every corner closing exactly. The Dehn invariant agrees: the Cleo-Icos's is not zero, so no single space-filler (a cube, a rhombohedron) could be its partner, and Antonius's is exactly −2 times the Cleo-Icos's, which the two-to-one ratio balances.

The same pieces build the **rhombic hexecontahedron**, known as 20 prolate golden rhombohedra round a point [Hafner and Žitko]: 20 Cleo-Icos round the centre (the icosahedron), 20 Antonius, and 20 Cleo-Icos at the tips (Figure 10c; checked against the rhombic hexecontahedron of our Bodhi Seed). The cut of a rhombohedron by those two planes is general (any parallelepiped splits into two corner tetrahedra of a sixth each and a middle of two thirds), and for the rhombohedron of 60° it gives the regular tetrahedra and octahedron of the tetrahedral–octahedral honeycomb. That the golden rhombohedron's corner is the icosahedron's central pyramid is known (Kabai: the "golden tetrahedron" is the tetrahedral part of a golden rhombohedron), and the prolate golden rhombohedron has been described as a distorted face-centred cubic cell whose cyclic twins, grouped as rhombic hexecontahedra, give icosahedral quasicrystals [Prodan et al. 2017]. What we add is the middle piece, Antonius, and with it the two-piece filling: the icosahedron's own pieces, with one partner, fill space. We found this split in none of the sources we read [Hafner and Žitko; Hafner and Komatar 2008; Kabai 2014; Prodan et al. 2017; Fang et al. 2013].

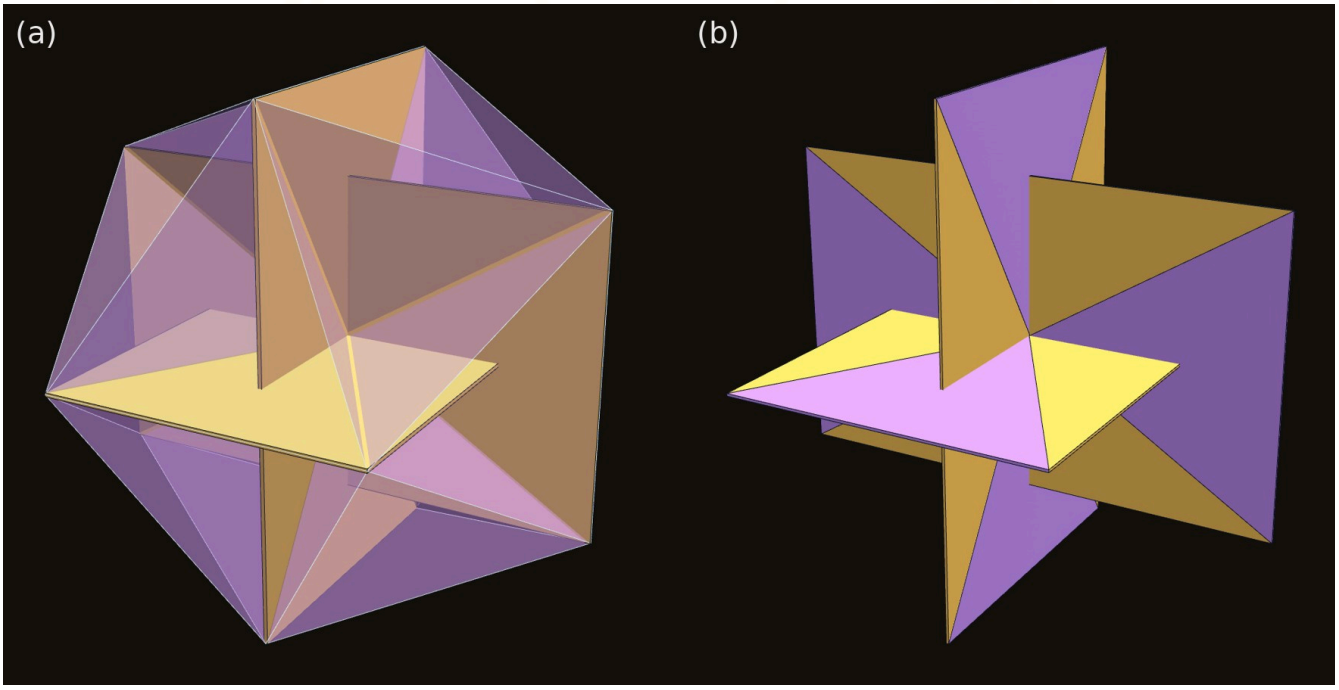


Figure 11. Both from the same viewpoint. (a) The icosahedron as 20 Cleo-Icos in glass, with Pacioli's three golden rectangles (gold) crossing at the centre, where the Cleo-Icos tips meet. (b) The rectangles alone, each cut by its diagonals: the triangles on the short sides (wisteria, 63.43° at the centre) are side faces of Cleo-Icos; those on the long sides (gold) have 116.57°.

Pacioli's golden rectangles. The three golden rectangles of Pacioli, at right angles through the centre, have their 12 corners at the icosahedron's 12 corners, the outer corners of the Cleo-Icos, and pass through the centre, where all 20 Cleo-Icos tips meet. Cut by its diagonals (two diameters), each rectangle falls into four triangles from the centre. The two on its short

sides, which are icosahedron edges, are exactly side faces of Cleo-Icos, walls between neighbours, with 63.43° at the centre; they are also halves of the golden rhombi that bound the golden rhombohedra above. The two on its long sides have 116.57° at the centre. 63.43° and 116.57° are the golden rhombus's two angles, so each rectangle is made of halves of golden rhombi, the same faces that bound the Cleo-Icos and the rhombohedra filled by Cleo-Icos and Antonius.

The octahedron, and the half Admiral Cap. Six Admiral Caps (our name for the two small tetrahedra that roof one of the icosahedron's six axis-parallel edges) turn the icosahedron into the regular octahedron [2]. Each cap splits into two **half caps**, and the octahedron is exactly **20 Cleo-Icos + 12 half caps** (volumes $\varphi^2/24$, $\varphi/24$ and $\varphi^6/6$ for an icosahedron of edge 1; sampling finds every point in exactly one piece). Along the edge it shares with the icosahedron the half cap's angle is 110.91°, Antonius's angle, so a Cleo-Icos and a half cap sharing a face make one **kite pyramid**, and the octahedron is also 8 Cleo-Icos + 12 kite pyramids. But Cleo-Icos and half caps (with or without Antonius) **cannot fill space**: the angles π, $\arccos(-\sqrt{5}/3)$ and $\arccos(1/\sqrt{3})$ are rationally independent, so there is an additive rule on angles that is zero on π and on every Cleo-Icos and Antonius angle and positive on the half cap's octahedral edges; round such an edge the angles could never total 2π. Space needs a further piece with tetrahedral angles, the regular tetrahedron, which gives back the octet [2].

6. What is known, and what is new

The base shape (the isogonal hexagon), the antiprism angles (listed for single antiprisms in the OEIS), the golden hexagon after Odom, the space-filling truncated tetrahedron, breathing pyrochlores and the Cleo-Icos itself (Mackay's unit tetrahedron) are all known, and are credited above. Our searches (14 October 2026: our reference library, web searches for the family, its base-angle formulas, flat-ended or isogonal hexagonal pyramids, and antiprism dihedral relations, and the OEIS) did not find the following. That is a report of what we searched, not a claim that they are unknown. The base-angle formulas are elementary, and the hexagons with alternating sides are standard in breathing kagome lattices; what we offer is the family as one object, and the links between its members:

- the family FUJI(r) of pyramids;
- the antiprism rule (Proposition 1) that links the family to the antiprisms, which combines the standard dihedral formula with the base-angle formula;
- that the equal-depth cube–octahedron solid has the cuboctahedral member's triangles exactly because its cube corners are that member's apexes (Section 5.4), and the wheel bound with the impossibility of five octagonal wheels (Section 5.5);
- the split of the prolate golden rhombohedron into Cleo-Icos + Antonius + Cleo-Icos, by which the icosahedron's central pyramid fills space with one partner, and the octahedron as 20 Cleo-Icos + 12 half Admiral Caps, which cannot fill space without a tetrahedral piece (Section 5.7);
- as a synthesis, the identifications that tie it together: FUJI's base as the golden hexagon after Odom, the breathing pyrochlore's hexagons as the family's bases, and the equal-depth cube–octahedron solid, whose triangles are the cuboctahedral member.

The family gives one shape and one parameter to things found separately:

- a gap piece in an icosahedral cluster;

- a space-filler found by computer packing;
- the hexagons of frustrated magnets;
- the faces of the equal-depth cube–octahedron solid;
- wheels of flat-ended triangles in the cuboctahedron, a sphere of twenty CLEOs, and six crossed circles that lead to the icosahedron.

In each, the parameter r is set by a different symmetry: five-fold for FUJI, cubic for the cube–octahedron solid, the breathing ratio for the pyrochlore.

Appendix: proofs

Base angles. The long side lies along a side of the big triangle (side $s = 1 + 2r$), so its distance from the centre is $s/(2\sqrt{3})$. The equilateral face on it has slant height $\sqrt{3}/2$, which gives $\cos L = (1 + 2r)/3$. A cut line lies at distance $s/\sqrt{3} - r\sqrt{3}/2 = (2 + r)/(2\sqrt{3})$, and the isosceles face on it has slant height $\sqrt{(1 - r^2/4)}$, which gives $\cos^2 S = (2 + r)/(3(2 - r))$.

Antiprism dihedral. With $c = \cos\theta$ and $s = \sin\theta$, unit side edges give $h^2 = (1 + 2c)/(8(1 + c))$. Writing $M = (R(1 + c)/2, Rs/2, 0)$:

- $T_1 - M = (R(4c + 3)(c - 1)/2, Rs(4c - 1)/2, h)$;
- $B_{-1} - M = (R(c - 1)/2, -3Rs/2, -h)$.

With $R^2 = 1/(4(1 - c)(1 + c))$, their dot product is

$$[(4c + 3)(1 - c) - 3(4c - 1)(1 + c) - 2(1 + 2c)] / (16(1 + c)) = -(4c - 1)/4.$$

So $\cos D_n = (1 - 4c)/3$, the standard antiprism dihedral. The numerical check, n = 3 to 12, agrees to 10^{-12} (study scripts and an independent review).

The cube–octahedron solid's member. Corner $A = (1, 1, 1)$; the triangle on $x + y + z = S$ has corners such as $(1, S - 1, 0)$ (after the cube cuts it) at distance $\sqrt{((2 - S)^2 + 1)}$ from A, the same for all six. For $3/2 < S < 2$, the long face A, $(1, S - 1, 0)$, $(1, 0, S - 1)$ lies in $x = 1$ and has the base edge $(S - 1)\sqrt{2}$; it is equilateral when $2(S - 1)^2 = (2 - S)^2 + 1$, i.e. $S^2 = 3$.

Credits

FUJI, Cleo-Icos, Antonius, the Lotus seed, the dictospheres, the DeltaZips and B1NR sculptures, the term "flat-ended triangle" and the names are DICTO's. Odom's work, the golden-hexagon construction after it, the space-filling truncated tetrahedron (Damasceno, Engel and Glotzer) and the breathing pyrochlore (Okamoto, Nilsen, Attfield and Hiroi) are credited to their authors. Data, scripts, proofs and checks: studies/2026-10-14-fuji-family studies/2026-10-14-icosa-cleo-pair and studies/2026-10-14-half-cap at https://github.com/DICTOR-Master/kaleidohedra.

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